

Hemant Kothari

hemantkothari1807@gmail.com | +91 7666346511 | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

SVKM's Dwarkadas J. Sanghvi College of Engineering, Mumbai, Maharashtra

Sept 2022 – July 2026

Bachelor of Technology in **Electronics and Communication Engineering**

GPGA: 8.30

Coursework: AIML, ML for Signal Processing, Computer Vision, Mathematics for ML

WORK EXPERIENCE

ML Intern – Bharti Airtel

Feb 2026 – June 2026

- Developed an XGBoost-based LTE customer churn prediction model for the Mumbai circle using customer and network KPIs, achieving **89.4%** accuracy, **84.1%** precision & **81.7%** recall.
- Developed **3 ML services** for KPI forecasting, clustering, and anomaly detection in Mumbai.
- Improved statistical and time-series pipelines for hourly network KPI analytics across Mumbai circle.
- Automated ETL & reporting workflows, reducing manual effort by **35+ hours/week** across 5 data analysts.

AI Developer Intern – Moksha Infotech

Oct 2025 – Present

- Processed and built FAISS vector DBs and Neo4j knowledge graphs for India's **financial laws and 70k+ cases** by scraping and cleaning legal corpora for semantic retrieval.
- Architected an **Agentic RAG** system to identify all applicable laws for a case and multi-step legal reasoning.
- Built **multi-agent** system generating citation-backed legal arguments using LLM tools & RAG pipelines.

Software Engineering & ML Intern – One Job

Sept 2025 – Nov 2025

- Enhanced semantic job-resume ranking using Sentence-BERT embeddings, increasing **Precision @10 by 23%**.
- Developed FastAPI inference services supporting 4,000+ low-latency prediction requests reliably.

PROJECTS

LLM Inference & RAG Experimentation Lab (LLM, Vector DB, Evaluation) [GitHub](#)

June 2026

- Architecting configurable RAG framework supporting **7 chunking strategies, 4 embeddings, and 4 vector databases**.
- Developing benchmarking platform integrating **20+ evaluation metrics** for reproducible LLM experimentation.
- Curated benchmark corpus spanning books, legal docs, technical docs, and Wikipedia.

Hybrid ViT-RAG Ophthalmology Intelligence System (ViT, PyTorch, LLM, RAG) [GitHub](#) | [Demo](#)

March 2026

- Fine-tuned **101M** parameters of RETFound ViT on **13k+ fundus images**, achieving **0.852 ROC-AUC & 0.843 QWK**.
- Built end-to-end pipeline converting retinal images into evidence-backed clinical decision reports.
- Combined 3 retinal datasets via feature engineering, improving external **ROC-AUC by 9%**.
- Built an **Agentic RAG** with hallucination / safety guardrails for clinical query answering.

FileSense (LLMs, PyQt, SQLite, Watchdog) [GitHub](#)

April 2026

- Built FileSense, an app that auto-generates file descriptions based on file type and content.
- Implemented real-time file monitoring and SQLite full-text search for low-latency indexing and retrieval.
- Designed entropy-based detection to flag sensitive data (API keys, credentials) before file sharing.

TECHNICAL SKILLS

- AI / ML & Data Science: LLMs, Agentic AI, RAG, NLP, Vector Databases, Knowledge Graphs, Recommender Systems, Neural Networks
- Libraries & Frameworks: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, FastAPI, REST API, Docker, AWS, Git
- Languages & Core Concepts: Python, Java, C++, SQL, NoSQL, DSA, OOP

ACHIEVEMENTS

Achievements:

- Secured **5th rank nationally out of 172 teams** at Codebot Hackathon, hosted by IIT Bombay's TIH at IISc Bangalore
- 4th position** at AgriTech Hackathon K. J. Somaiya College of Engineering